

Calculus II

Trigonometry review

Todor Milev

2019

Outline

- 1 Review of trigonometry
 - The Trigonometric Functions
 - Trigonometric Identities
 - Trigonometric Identities and Complex Numbers
 - Graphs of the Trigonometric Functions

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- 1 Review of trigonometry
 - The Trigonometric Functions
 - Trigonometric Identities
 - Trigonometric Identities and Complex Numbers
 - Graphs of the Trigonometric Functions
- 2 Inverse Trigonometric Functions

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Geometric interpretation of all trigonometric functions

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Definition (Trigonometric Identity)

A trigonometric identity is an equality between the trigonometric functions in one or more variables that holds for all values of the involved variables in the domains of all of the expressions.

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- By convention, when dealing with trigonometric identities we do not account for the domains of the involved expressions.
- For example, $\frac{\sin \theta}{\sin \theta} = 1$ is considered a valid trigonometric identity, although, when considered as a function, the left hand side is not defined for $\theta \neq 0$.

$$\begin{aligned}\sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y}\end{aligned}$$

- $\csc \theta = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
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$$\sin^2 \theta + \cos^2 \theta$$

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$$\begin{aligned}& \sin^2 \theta + \cos^2 \theta \\&= \frac{y^2}{r^2} + \frac{x^2}{r^2} \\&= \frac{y^2 + x^2}{r^2} \\&= \frac{r^2}{r^2} \\&= 1\end{aligned}$$

Therefore $\sin^2 \theta + \cos^2 \theta = 1$.

$$\begin{array}{ll} \sin \theta = \frac{y}{r} & \csc \theta = \frac{r}{y} \\ \cos \theta = \frac{x}{r} & \sec \theta = \frac{r}{x} \\ \tan \theta = \frac{y}{x} & \cot \theta = \frac{x}{y} \end{array}$$

Example ($\tan^2 \theta + 1 = \sec^2 \theta$)

Prove the identity
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The remaining identities are consequences of the addition formulas:

$$\begin{aligned}\sin(x + y) &= \sin x \cos y + \cos x \sin y \\ \cos(x + y) &= \cos x \cos y - \sin x \sin y\end{aligned}$$

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Substitute $-y$ for y , and use the fact that $\sin(-y) = -\sin y$ and $\cos(-y) = \cos y$:

$$\begin{aligned}\sin(x - y) &= \sin x \cos y - \cos x \sin y \\ \cos(x - y) &= \cos x \cos y + \sin x \sin y\end{aligned}$$

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To get the half-angle formulas, solve these equations for $\cos^2 x$ and $\sin^2 x$ respectively.

$$\cos^2 x = \frac{1 + \cos(2x)}{2}, \quad \sin^2 x = \frac{1 - \cos(2x)}{2}$$

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Divide the first equation by the second, and then cancel $\cos x \cos y$ from the top and bottom:

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

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$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Do the same for the subtraction formulas:

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

- Positive angles are obtained by rotating counterclockwise.

$$\begin{array}{ll} \sin \theta = \frac{y}{r} & \csc \theta = \frac{r}{y} \\ \cos \theta = \frac{x}{r} & \sec \theta = \frac{r}{x} \\ \tan \theta = \frac{y}{x} & \cot \theta = \frac{x}{y} \end{array}$$

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- If (x, y) is on the terminal arm of the angle θ , then $(x, -y)$ is on the terminal arm of $-\theta$.

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- $\sin$ is an **odd function**.

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- $\sin$ is an odd function.
- $\cos$ is an even function.

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- 2π represents a full rotation.

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- We say $\sin$ and $\cos$ are 2π -periodic.

Definition (Complex numbers)

The set of complex numbers $\mathbb{C}$ is defined as the set

$$\{a + bi \mid a, b - \text{real numbers}\},$$

where the number i is a number for which

$$i^2 = -1 \quad .$$

The number i is called the imaginary unit. .

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$$\begin{aligned}(a + bi)(c + di) &= ac + adi + bci + bdi^2 = \textcolor{red}{ac} + adi + bci - \textcolor{red}{bd} \\ &= (\textcolor{red}{ac} - \textcolor{red}{bd}) + i(ad + bc)\end{aligned}$$

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Euler's Formula

Theorem (Euler's Formula)

$$e^{ix} = \cos x + i \sin x,$$

where $e \approx 2.71828$ is Euler's/Napier's constant .

Proof.

Recall $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1) \cdot n$. Borrow from Calc II the f-las:



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$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^{2n}}{(2n)!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$$

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Rearrange.



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Rearrange. **Plug-in** $z = ix$.



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$$e^{ix} = 1 + ix - \frac{x^2}{2!} - i \frac{x^3}{3!} + \frac{x^4}{4!} + i \frac{x^5}{5!} - \dots$$

Rearrange. Plug-in $z = ix$. Use $i^2 = -1$.



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Rearrange. Plug-in $z = ix$. Use $i^2 = -1$. Multiply $\sin x$ by i .



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Rearrange. Plug-in $z = ix$. Use $i^2 = -1$. Multiply $\sin x$ by i . **Add to get**
 $e^{ix} = \cos x + i \sin x.$



Trigonometric Identities Revisited

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All trigonometric formulas can be easily derived using the above formulas.

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Express $\sin(3x)$ and $\cos(3x)$ via $\cos x$ and $\sin x$.

- Recall Euler's formula: $e^{i\alpha} = \cos \alpha + i \sin \alpha$.

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The real parts of the starting and final expression must be equal; likewise the imaginary parts must be equal; therefore:

$$\begin{aligned}
 \cos(3x) &= \cos^3 x - 3\cos x \sin^2 x \\
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- If we translate the graph of $\cos x$ by $\frac{\pi}{2}$ units to the right we get the graph of $\sin x$. This is a consequence of $\cos\left(x - \frac{\pi}{2}\right) = \sin x$.

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- We call it $\arcsin$ or $\sin^{-1}$.
- $\arcsin x = y \Leftrightarrow \sin y = x$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

Example

Find $\arcsin\left(\frac{1}{2}\right)$.

Observation

- $\arcsin y =$ *the appropriate angle whose sine equals y .*

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- $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}.$

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- *Important: the output angle must lie in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$.*

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- $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$.
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- Therefore $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$.

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- Therefore $\arcsin(\sin 1.5) = 1.5$.

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$$\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1.$$

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Then $\sin y = x$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.

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Theorem (The Derivative of $\arcsin x$)

$$\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1.$$

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Important facts about arcsin:

- 1 Domain: ?
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- 3 $\arcsin x = y \Leftrightarrow \sin y = x$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
- 4 $\arcsin(\sin x) = x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.
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(The proof is similar to the proof of the formula for the derivative of $\arcsin x$.)

Example

Rewrite $\sin(2 \arccos(x))$ as an algebraic expression of x and $\sqrt{1 - x^2}$.

$$\sin(2 \arccos(x))$$

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Rewrite $\sin(2 \arccos(x))$ as an algebraic expression of x and $\sqrt{1 - x^2}$.
To simplify $\arccos x$ we try to use $\cos(\arccos x) = x$.

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$$\sin(2 \arccos(x)) = \sin(2y)$$

$$| \text{ Set } y = \arccos x$$

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$$\begin{aligned}\sin(2 \arccos(x)) &= \sin(2y) \\ &= ?\end{aligned}$$

Set $y = \arccos x$
Express via $\sin y, \cos y$

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Rewrite $\sin(2 \arccos(x))$ as an algebraic expression of x and $\sqrt{1 - x^2}$. To simplify $\arccos x$ we try to use $\cos(\arccos x) = x$. Therefore our aim is to **rewrite the expression only using the cos function**.

$$\begin{aligned}\sin(2 \arccos(x)) &= \sin(2y) \\ &= 2 \cos y \sin y \\ &= 2 \cos y \left(\pm \sqrt{1 - \cos^2 y} \right)\end{aligned} \quad \left| \begin{array}{l} \text{Set } y = \arccos x \\ \text{Express via } \sin y, \cos y \\ \text{Express } \sin y \text{ via } \cos y \end{array} \right.$$

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Rewrite $\cos(3 \arccos(x))$ as an algebraic expression of x and $\sqrt{1 - x^2}$.

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$$\cos(3 \arccos(x)) = \cos(3y) \quad \Big| \quad y = \arccos x$$

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$y = \arccos x$
Angle sum f-la

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$$\begin{aligned}\cos(3 \arccos(x)) &= \cos(3y) = \cos(2y + y) \\ &= \cos(2y) \cos y - \sin(2y) \sin y\end{aligned} \quad \left| \begin{array}{l} y = \arccos x \\ \text{Angle sum f-la} \end{array} \right.$$

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 &= (\text{?}) \cos y \\
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$y = \arccos x$
 Angle sum f-la
 Express via
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Express $\sin y$
 via $\cos y$

$x = \cos y$

Example

Rewrite $\cos(3 \arccos(x))$ as an algebraic expression of x and $\sqrt{1-x^2}$.

To simplify $\arccos x$ we try to use $\cos(\arccos x) = x$. Therefore our aim is to rewrite the expression only using the $\cos$ function.

$$\begin{aligned}
 \cos(3 \arccos(x)) &= \cos(3y) = \cos(2y + y) & y &= \arccos x \\
 &= \cos(2y) \cos y - \sin(2y) \sin y & \text{Angle sum f-la} \\
 &= (\cos^2 y - \sin^2 y) \cos y & \text{Express via} \\
 &\quad - 2 \sin y \cos y \sin y & \sin y, \cos y \\
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 &= \cos^3 y - 3 \sin^2 y \cos y & \text{Express } \sin y \\
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- Draw a right triangle with opposite x and adjacent 1.
- Length of hypotenuse = $\sqrt{1^2 + x^2}$.
- Then $\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}$.

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Evaluate

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The remaining inverse trigonometric functions aren't used as often:

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Table of derivatives of inverse trigonometric functions:

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

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$$\frac{d}{dx}(\operatorname{arccsc} x) = -\frac{1}{x\sqrt{x^2-1}}$$

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$$= - \frac{1}{(\arcsin x)^2 \sqrt{1-x^2}}.$$

All of the inverse trigonometric derivatives also give rise to integration formulas. These two are the most important:

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C.$$

$$\int \frac{1}{x^2+1} dx = \arctan x + C.$$